The Daniel Test

This Test aims to be a general assessment of the software and the reader performance. It should ideally be executed for each major Software update. It measures the performance by grading Noise, SNR, Spikes and anomalies metrics for each mologram and then for the whole chip, which gives a fail/pass note for each mologram and respectively the whole chip.

Note: it is best to use this test with the same chip, dedicated for this test.

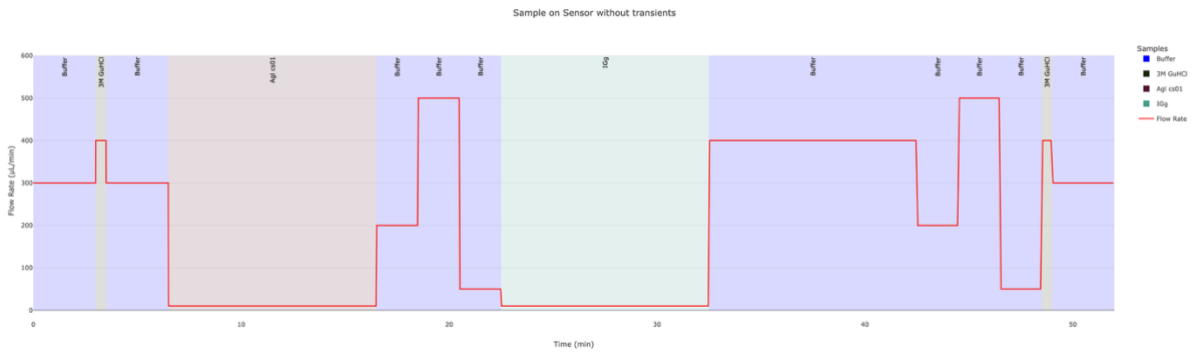
IMPORTANT - The name of the experiment downloaded results csv file must start with:

"YYYY-MM-DD_HH:MM:SS_THE_DANIEL_TEST_{reader name} …."

The experiment

$$Regeneration(GuHCl) \rightarrow Ligand\ 1\ (protein\ AGL\ cs01)X \rightarrow Analyte\ (IgG)X \rightarrow Slow\ Dissociation \rightarrow buffer\ \frac{400\mu l}{min} \rightarrow Regeneration\ (GuHCl)$$

$$X = Flow\ rate\ change\ in\ the\ high\ signal\ part$$



Custom Phase

Type	Sample	Vial	Duration	Pickup	Flowrate
Custom Injections					
Baseline	Buffer		3 min		300 uL/min
Regeneration	3M GuHCl	1-A1	0.5 min	200 µL	400 uL/min
Baseline	Buffer		3 min		300 uL/min
Association	Agl cs01	1-A2	10 min	100 µL	10 uL/min
Baseline	Buffer		2 min		200 uL/min
Baseline	Buffer		2 min		500 uL/min
Baseline	Buffer		2 min		50 uL/min
Association	IGg	1-A3	10 min	100 µL	10 uL/min
Dissociation	Buffer		10 min		400 uL/min
Baseline	Buffer		2 min		200 uL/min
Baseline	Buffer		2 min		500 uL/min
Baseline	Buffer		2 min		50 uL/min
Regeneration	3M GuHCl	1-A1	0.5 min	200 µL	400 uL/min
Baseline	Buffer		3 min		300 uL/min

New Group

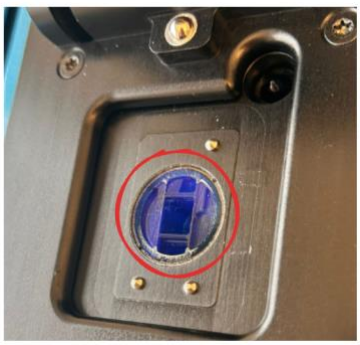
+ Add Command

Export Settings

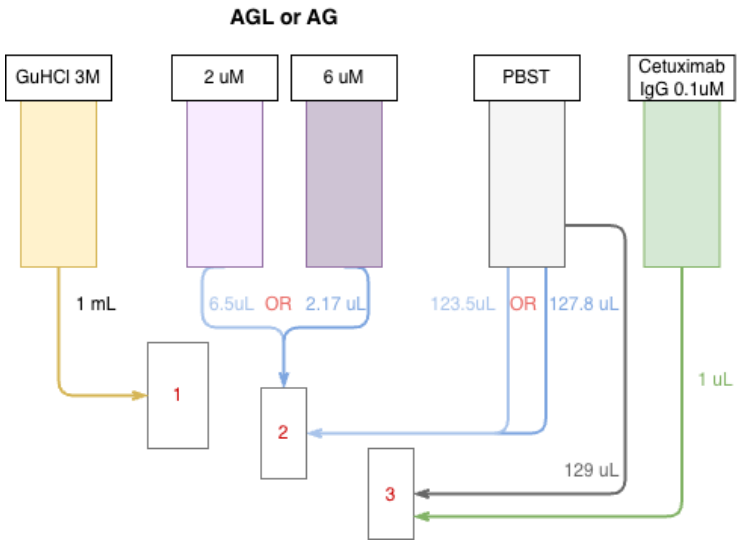
- [ ] Export Fluorescence Images
  Every n steps: 100
- [x] Export Surface Images
  Every n steps: 100
- [x] Export Result Images
  Every n steps: 100

Below are the technical steps on how to conduct it:

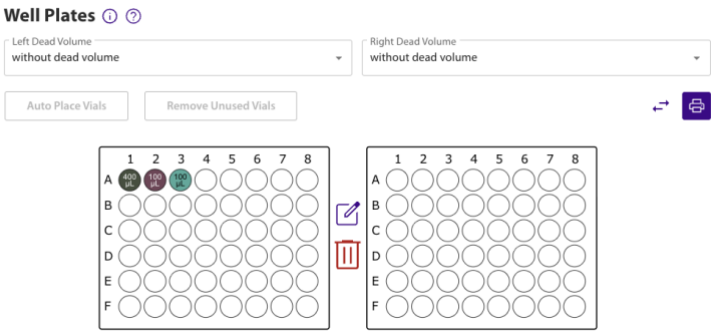
1. Use a regular chip with molograms 8x8 oligo1-oligo2-PEG
2. Use an assembly holder with multi-channel configuration
3. Screw tight the chip to the holder assembly
4. Clean the glass surface of the reader

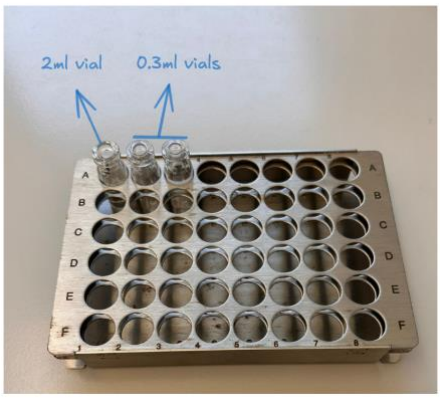


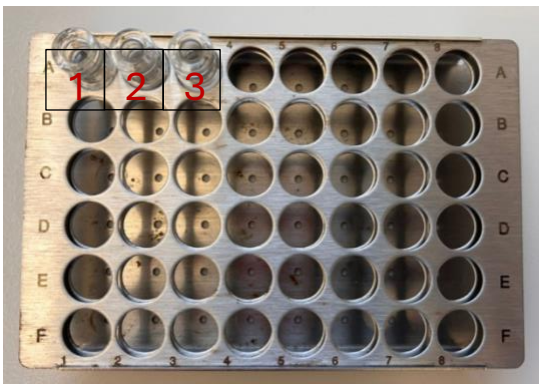
5. Fill bottles of buffer and water of autosampler, if are not filled.
6. Prepare the samples for the experiment



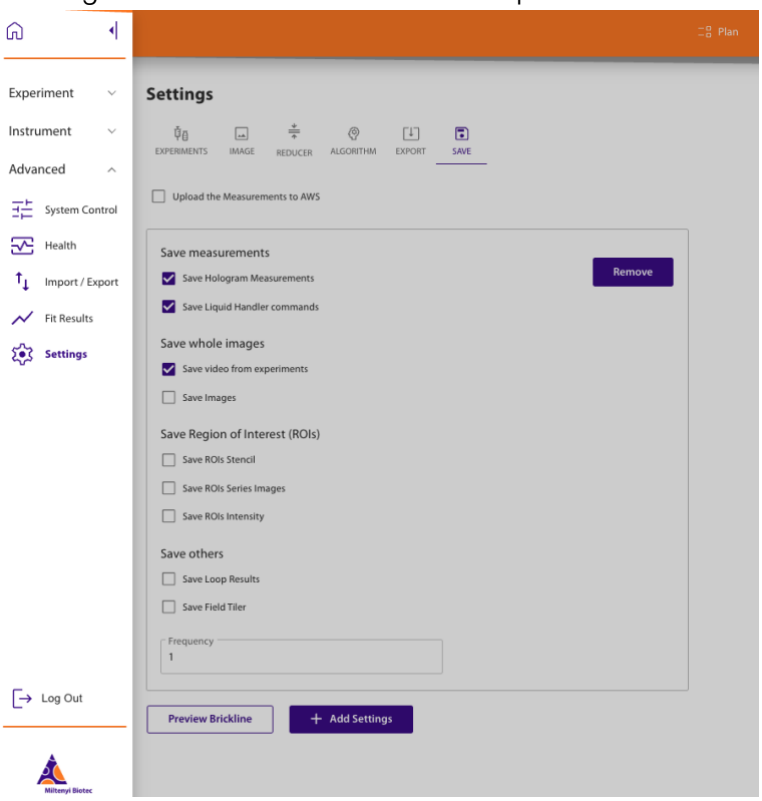
7. Fill the vials for this experiment



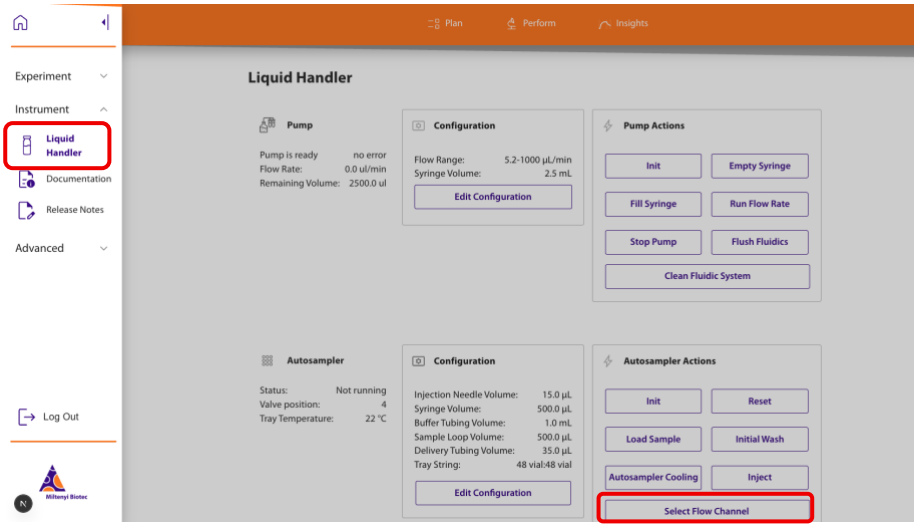


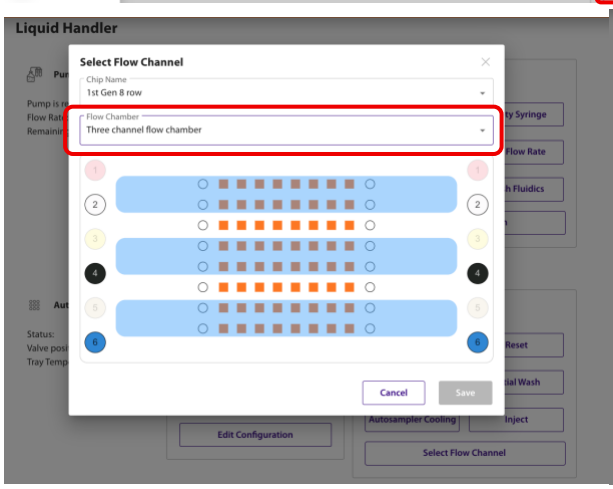


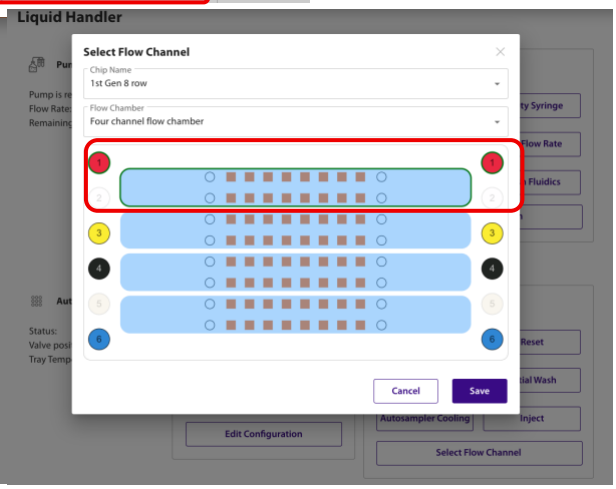
8. Settings – SAVE – Save video from experiments

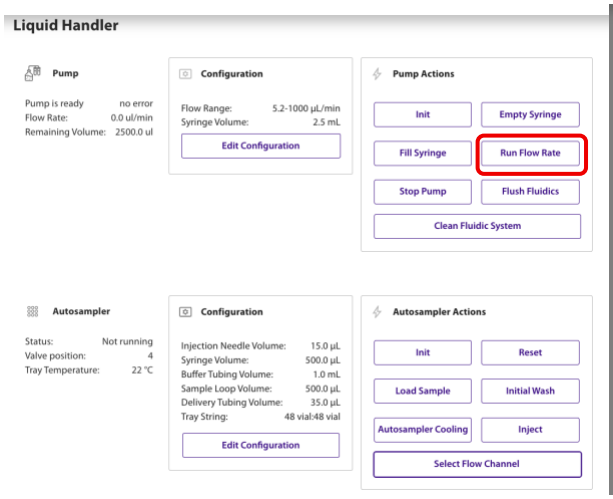


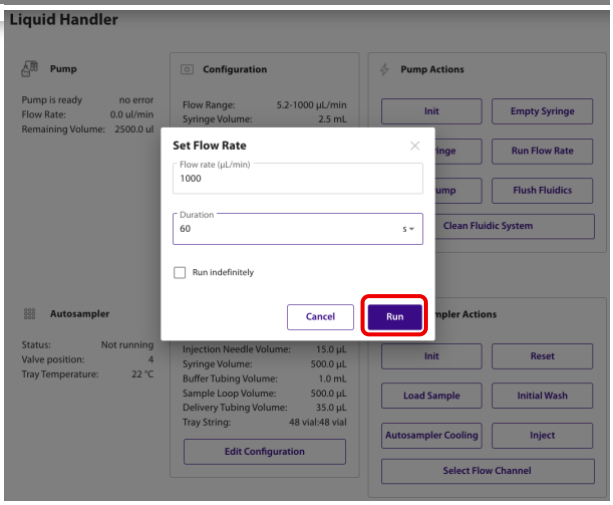
9. Fill all the unused channels with the buffer solution, make sure to end this step on the channel for which the entry is the same as the experiment one (usually 4). Below are examples with 3 and 4 channels configurations


Plan
Perform
Insights
Experiment
Instrument
Liquid Handler
Documentation
Release Notes
Advanced
Log Out
Liquid Handler
Pump
Pump is ready no error
Flow Rate: 0.0 ul/min
Remaining Volume: 2500.0 ul
Configuration
Flow Range: 5.2-1000 µL/min
Syringe Volume: 2.5 mL
Edit Configuration
Pump Actions
Init
Empty Syringe
Fill Syringe
Run Flow Rate
Stop Pump
Flush Fluidics
Clean Fluidic System
Autosampler
Status: Not running
Valve position: 4
Tray Temperature: 22 °C
Configuration
Injection Needle Volume: 15.0 µL
Syringe Volume: 500.0 µL
Buffer Tubing Volume: 1.0 mL
Sample Loop Volume: 500.0 µL
Delivery Tubing Volume: 35.0 µL
Tray String: 48 vial:48 vial
Edit Configuration
Autosampler Actions
Init
Reset
Load Sample
Initial Wash
Autosampler Cooling
Inject
Select Flow Channel

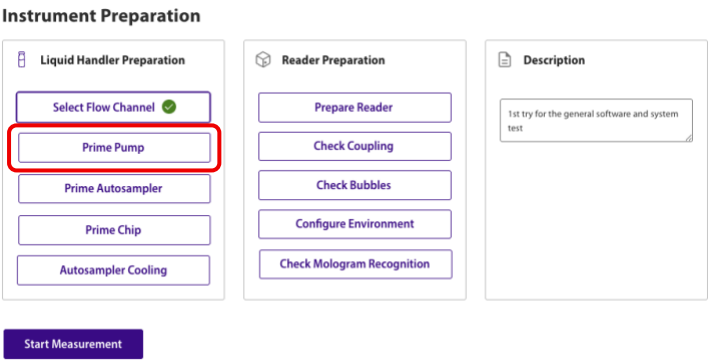

Liquid Handler
Select Flow Channel
Chip Name
1st Gen 8 row
Flow Chamber
Three channel flow chamber
Cancel
Save


Liquid Handler
Select Flow Channel
Chip Name
1st Gen 8 row
Flow Chamber
Four channel flow chamber
Cancel
Save

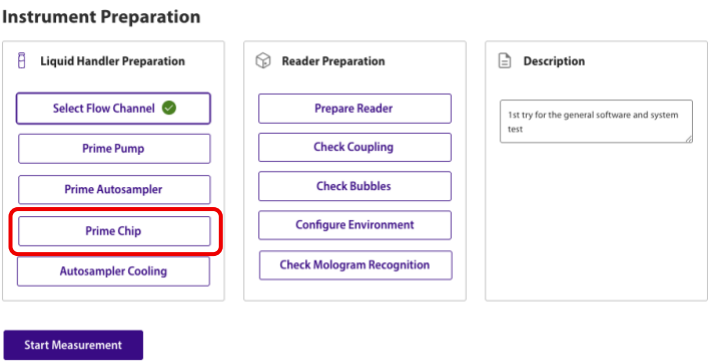

Liquid Handler
Pump
Pump is ready no error
Flow Rate: 0.0 ul/min
Remaining Volume: 2500.0 ul
Configuration
Flow Range: 5.2-1000 µL/min
Syringe Volume: 2.5 mL
Edit Configuration
Pump Actions
Init
Empty Syringe
Fill Syringe
Run Flow Rate
Stop Pump
Flush Fluidics
Clean Fluidic System
Autosampler
Status: Not running
Valve position: 4
Tray Temperature: 22 °C
Configuration
Injection Needle Volume: 15.0 µL
Syringe Volume: 500.0 µL
Buffer Tubing Volume: 1.0 mL
Sample Loop Volume: 500.0 µL
Delivery Tubing Volume: 35.0 µL
Tray String: 48 vial:48 vial
Edit Configuration
Autosampler Actions
Init
Reset
Load Sample
Initial Wash
Autosampler Cooling
Inject
Select Flow Channel

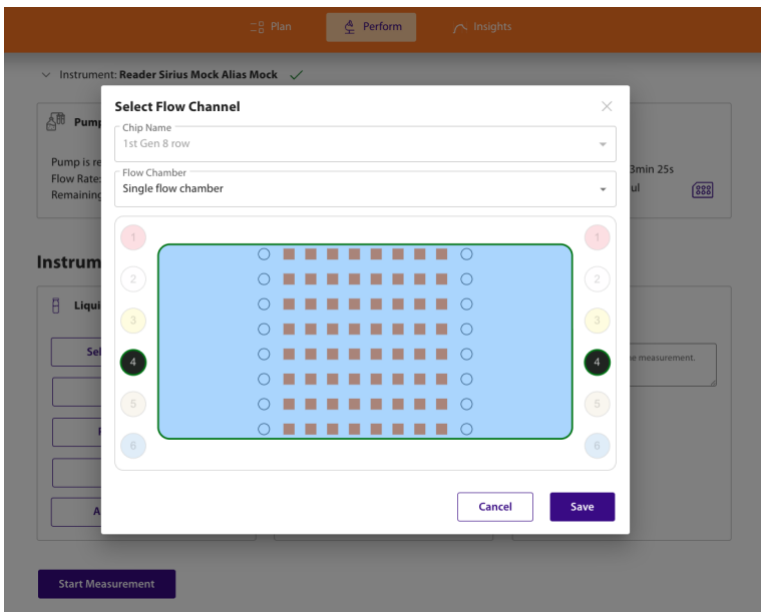

Liquid Handler
Set Flow Rate
Flow rate (µL/min)
1000
Duration
60
s
Run indefinitely
Cancel
Run

10. Prime pump

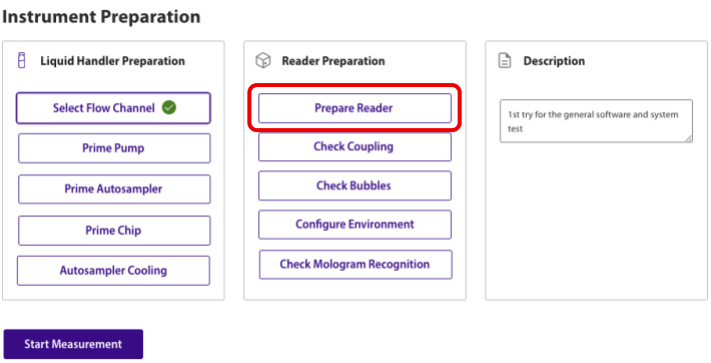


11. Prime chip

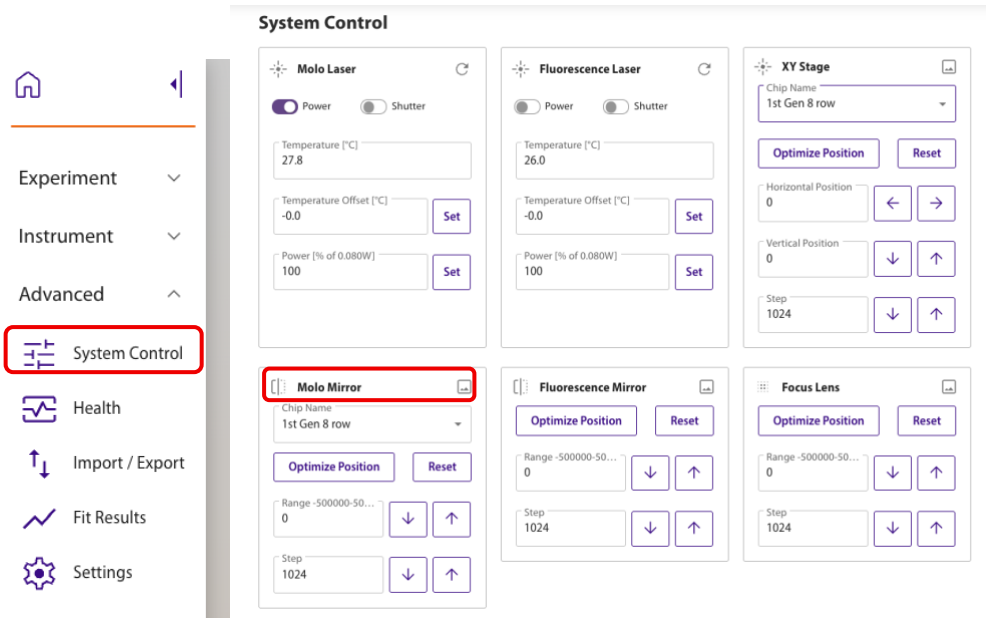




12. Prepare reader – make sure it passes all



13. System controls – verify values and adjust



14. Move out of good position for molo (>40k steps), XY (100k steps), focus.
15. Prepare reader again – make sure it passes all
16. Check bubbles – Check that the surface image looks good and that there are no bubbles
17. Check coupling – Check the coupling power profile, intensity and mologram recognition
18. Start the measurement "Daniel General Test"
19. Look at logs and for issues
20. Check metrics from the data: Noise, SNR, Spikes, Anomalies (steps), RMSE to desired model plot ("perfect plot").
21. Analyze the surface images and the result images
22. Check the video of the experiment by running the following (download it from the reader): scp -r container@READER:/home/container/lino_data/experiment_test_data/YYYY-MM-DD/{name of your experiment} {name of the arrival directory}